
GAN-Based Synthetic Data Augmentation to Address Class Imbalance in RetinaMNIST Dataset

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Abstract

Medical imaging datasets often suffer from significant class imbalance, limiting the effectiveness of deep learning models in identifying rare conditions. In this study, we focus on the RetinaMNIST dataset, a benchmark for retinal disease classification, where minority classes are underrepresented. To address this, we trained a Generative Adversarial Network (GAN) enhanced with residual connections on samples from the minority Class 4 to generate synthetic retinal images. These generated images were combined with the original training set to upsample the minority class. A VGG19-based Convolutional Neural Network (CNN) was trained on both the original and augmented datasets to assess performance improvements. Results show that augmenting the training data with GAN-generated samples significantly improves minority class recall and overall model robustness, highlighting the potential of residual GAN architectures and deep feature extractors like VGG19 in medical image augmentation tasks.

1 Introduction

Deep learning models have achieved remarkable success in a wide range of computer-vision applications, including medical-image analysis [2,32]. However, a major bottleneck in deploying such models in healthcare is the scarcity of large, diverse, and balanced datasets [2,3]. In medical data, minority classes often represent critical yet rare conditions, and their under-representation leads to models biased toward frequent classes and poor sensitivity for rare diseases—a serious limitation in clinical decision-making [3,30].

The RetinaMNIST dataset, composed of retinal fundus images categorized into five classes, exemplifies this challenge: Class 4 instances are severely under-represented compared with other categories [1]. Traditional data-augmentation techniques (e.g. flips or rotations) provide limited diversity and do not fully correct such severe imbalance [14]. Generative Adversarial Networks (GANs) offer a promising alternative by learning the data distribution and generating realistic synthetic samples [8].

In this project, a GAN architecture enhanced with residual connections [9] was trained specifically on Class 4 images from RetinaMNIST. The resulting synthetic samples were used to up-sample Class 4, augmenting the original training set. A VGG19-based Convolutional Neural Network [5] was then trained separately on the original and augmented datasets. Performance was evaluated with Accuracy, Precision, Recall, and F1-Score. The results highlight the effectiveness of residual-based GANs for synthetic augmentation and the robustness of deep feature extractors such as VGG19 in medical-imaging tasks, offering a pathway to mitigate class-imbalance challenges in clinical datasets.

2 Related Work

2.1 GANs in medical imaging

Generative Adversarial Networks (GANs) are widely used in medical-image analysis for data-set augmentation, image-to-image translation, and domain adaptation [15]. The survey of Yi *et al.* [2] concludes that GAN-based synthesis can mitigate data scarcity when image realism is carefully validated. Building on that survey, Skandarani *et al.* [27] benchmarked DCGAN, StyleGAN, and later style-based models on cine-MRI, liver CT, and RGB-retina data, showing that only a subset of GANs achieve both low Fréchet Inception Distance (FID) scores and downstream performance gains.

2.2 Class imbalance

Severe class imbalance still limits medical-image classifiers. Buda *et al.* [3] demonstrated that minority-class recall drops sharply in CNNs trained on skewed data. Johnson and Khoshgoftaar’s survey [30] reviews reweighting, random over/undersampling, and synthetic-minority generation, while Gupta *et al.* [31] propose complexity-aware sampling to select informative minority images.

2.3 RetinaMNIST focus

RetinaMNIST, part of the MedMNIST v2 benchmark [1], provides 64×64 multi-label retinal-disease images with a pronounced imbalance—Class 4 has the fewest examples. Current RetinaMNIST studies focus on transfer or meta-learning rather than data-level augmentation: Ma *et al.* [28] benchmarked ImageNet-initialised ResNet and EfficientNet, while Goyal *et al.* [29] explored few-shot meta-learning.

2.4 Our contribution

We introduce a GAN-driven augmentation pipeline that selectively enriches the under-represented Class 4. By coupling high-quality synthetic samples with traditional flips and rotations, we test whether data-level balancing improves multi-label performance on RetinaMNIST beyond what transfer learning alone can achieve.

3 Methods

3.1 Base CNN Architecture

We propose a hybrid Convolutional Neural Network architecture that leverages the representational strength of pre-trained VGG19 model [5] coupled with custom built Dense Neural Network (DNN) classifier. We used VGG19 model, by leveraging the transfer learning, specifically to grasp high-level feature representations of the train images [16]. The last 15 layers of the VGG model is set to trainable along with the DNN. The input to the CNN model is an image of dimensions $64 \times 64 \times 3$ RGB images. VGG19 model followed by GlobalMaxPooling layer is used to flatten spatial representations. Three dense layers with 110, 90, 22 nodes respectively, all with ReLU activation functions [19] are used. To aid for better regularization, dropout layer [13] is added with deactivation probability of 0.1. The output layer of the base CNN model has five nodes with *softmax* activation function for multi-class classification. As considered for the optimization, Adaptive Moment Estimation (Adam) optimizer is used [10] along with clip norm value set to 1.0 for better gradient stability. Categorical Crossentropy loss function is used in this model. To prevent the model from overfitting, the regularization of L2 with $\lambda = 2e^{-3}$ is used [11]. The model is evaluated using class-aware metrics including precision, recall, f1-score and accuracy to account for class imbalance [4]. A total of 20,092,921 parameters are embedded into the base CNN model, where 19,832,761 are trainable parameters and 260,160 are non trainable parameters.

3.2 GAN Architecture

The generator is a residual upsampling network that projects a 128-dimensional random noise vector into a $4 \times 4 \times 512$ feature map, followed by a sequence of four residual blocks to progressively

upsample the resolution to $64 \times 64 \times 3$. Each residual block consists of a Conv2DTranspose layer for upsampling, batch normalization, ReLU activation, and a Conv2D layer, with a skip connection implemented using a 1×1 Conv2DTranspose operation to preserve feature flow [17]. The final output layer applies a convolution followed by a *Tanh* activation function to produce an RGB image with pixel values scaled between $[-1, 1]$. The generator comprises 3,610,211 parameters, of which 3,591,907 are trainable and 18,304 are non-trainable.

The discriminator is a deep convolutional network that processes $64 \times 64 \times 3$ images through four Conv2D layers with increasing filters (96, 54, 48, and 18), each followed by batch normalization, LeakyReLU activation ($\alpha = 0.2$), and dropout regularization [13]. The extracted features are flattened and passed through a fully connected layer with 50 neurons (ReLU activation), and finally through a dense layer outputting a single real/fake score. The GAN was trained using hinge loss for both generator and discriminator [21], with the discriminator updated four times per generator update to maintain training stability. Adam optimizers were used with different learning rates (3×10^{-4} for the generator and 3×10^{-5} for the discriminator), and learning rates were scheduled using polynomial decay [20]. Gradient clipping [22] was applied to discriminator updates, and synthetic images were saved at the end of each epoch using a custom monitoring callback. The discriminator contains 2,864,513 parameters, of which 2,864,081 are trainable and 432 are non-trainable. Training was conducted for 300 epochs on the augmented RetinaMNIST dataset, targeting the generation of high-quality synthetic samples to upsample the minority class.

3.3 Synthetic Image Filtering

To ensure only high-quality synthetic images were used for data augmentation, a custom image filtering strategy was applied. First, the Structural Similarity Index Measure (SSIM) was computed between each generated image and all real Class 4 images from the RetinaMNIST dataset [23]. For each synthetic image, the highest SSIM score obtained against any real image was recorded. The synthetic images were then ranked based on these maximum SSIM scores. The top-300 synthetic images with the highest similarity to real samples were selected for augmentation. This SSIM-based filtering [26] ensured that only images closely resembling real pathological patterns were included. Although classifier-based confidence thresholds were initially considered, the primary selection criterion for this study was the SSIM score ranking, allowing a purely visual similarity-driven selection of synthetic samples.

3.4 Augmentation Strategy

From the top-300, 236 randomly selected synthetic images were combined with the original RetinaMNIST training dataset to augment the minority Class 4 category [1]. Each class in the train dataset consists of 500 samples including previously augmented instances. Whereas, initially, Class 4 samples are only 264 in count even after traditional data augmentation [14]. Each synthetic image was labeled as Class 4 and resized to 64×64 pixels to match the input requirements of the classifier. After normalization, the synthetic samples were appended to the real training data, effectively increasing the number of Class 4 examples and partially balancing the class distribution [7]. No synthetic images were added to other classes, ensuring targeted augmentation focused solely on improving minority class representation. The augmented dataset was then shuffled and prepared as a TensorFlow dataset for retraining the classifier.

4 Experiments

4.1 Dataset (RetinaMNIST)

RetinaMNIST is a part of the MedMNIST v2 collection, a benchmark suite designed for lightweight biomedical image classification tasks [1]. Specifically, RetinaMNIST consists of retinal fundus photographs categorized into five classes corresponding to different retinal disease conditions. The images are preprocessed and resized to 64×64 pixels to facilitate efficient model training. Originally derived from a clinical dataset, RetinaMNIST retains essential pathological features while simplifying input size for deep learning experiments. It presents a challenging setting due to significant class imbalance, with certain disease categories heavily under-represented compared to others [3]. The

dataset is split into training, validation, and test sets and is widely used for benchmarking medical image classification, few-shot learning, and synthetic data augmentation methods [29].

The class distribution of train dataset is as follows

Class	Count
Class 0	486
Class 1	128
Class 2	206
Class 3	94
Class 4	66

Table 1: Class distribution of RetinaMNIST train dataset

Class 4 is significantly underrepresented in the dataset, with only 66 available samples. This limited sample size is insufficient for a deep learning model to learn the full distribution of class-specific features, especially in high-dimensional image space. Consequently, the focus of this work is directed toward the targeted generation of synthetic class 4 samples to mitigate this imbalance and enhance the model’s ability to accurately classify this minority class.

4.2 Classifier Setup

For evaluating the effect of synthetic data augmentation, a VGG19-based Convolutional Neural Network model (CNN), pretrained on ImageNet [5,6] was used. The network consists of multiple convolutional blocks with increasing filter sizes, each followed by ReLU activation and max pooling, and a final set of fully connected layers leading to a softmax output over five classes [19]. Images were resized to 64×64 pixels and normalized to have pixel values between 0 and 1 before feeding into the model. The classifier was trained for 19 epochs using the Adam optimizer [10] with a batch size of 128, and the loss function used was categorical cross-entropy [11]. Model performance was evaluated based on Accuracy, Precision, Recall, F1-Score, and Area Under the ROC Curve (AUC) on the validation and test sets [4].

4.3 Generative Adversarial Network Setup

We design a Generative Adversarial Network (GAN) with a ResNet-style generator [17] and a conventional Convolutional Neural Network (CNN) discriminator. The generator incorporates skip connections [9] to facilitate effective gradient flow and enhance feature representation learning. It takes a latent vector of dimension 128 as input and generates synthetic images. The architecture consists of four residual upsampling blocks with 256, 128, 64, and 32 filters respectively, each using a kernel size of 3. The final layer outputs an RGB image with a *Tanh* activation function.

The discriminator is a sequential CNN model with input dimensions of $64 \times 64 \times 3$, consisting of four convolutional blocks with varying filter and kernel sizes, followed by a two-layer Dense Neural Network (DNN) with 50 and 1 units, respectively. LeakyReLU is used as the activation function for intermediate layers, while no activation is applied at the output [13,19].

The generator is trained with a learning rate of 2.8×10^{-4} , while the discriminator uses a lower learning rate of 3×10^{-5} to prevent early domination over the generator during training [10]. A polynomial decay scheduler is employed for both optimizers, with decay steps of 5000 for the generator and 4800 for the discriminator [20]. The final learning rates are 9×10^{-5} and 8×10^{-7} for the generator and discriminator, respectively. Adam optimizers are used with $\beta_1 = 0.0$ and $\beta_2 = 0.999$ for both generator and discriminator.

Instead of the standard Binary Crossentropy, hinge loss is adopted for both models to mitigate discriminator overfitting and avoid vanishing gradients [18,21]. Additionally, Structural Similarity Index Measure (SSIM) loss is integrated into the generator loss to enhance perceptual quality [23]. The final generator loss combines hinge loss with SSIM loss, weighted by a factor of 0.3. The model is trained for 300 epochs.

$$\text{Generator Loss: } \mathcal{L}_G = \mathcal{L}_{\text{Hinge}} + 0.3 \times \mathcal{L}_{\text{SSIM}}$$

$$\text{Discriminator Loss: } \mathcal{L}_D = \mathcal{L}_{\text{Hinge}}$$

4.4 Training

First, a baseline performance metrics are established by training the base CNN classifier model using original dataset. This dataset is upsampled to ensure each class category has 500 instances in train dataset. The class 4 samples are oversampled to reach the 500 instance limit to give fair advantage for each class for the classifier to learn.

Next, a GAN model is trained till 300 epochs purely on class 4 images to generate 500 synthetic class 4 images. Of all the generated images, top 300 images are sorted based on SSIM score. From this set of synthetic images, 236 images are augmented to the original dataset which includes upsampled instances. The oversampled class 4 instances are purged and randomly picked 236 samples are augmented instead.

Then the base CNN classifier model, with hyper parameters kept intact, is again trained on this newly augmented train dataset to establish new performance metrics.

4.4.1 Training GAN

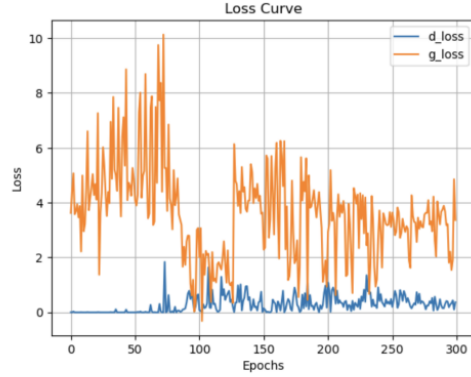


Figure 1: Plot of loss curves of generator with resnet-style blocks and discriminator

Figure 2 illustrates the training dynamics of the GAN model, with generator loss shown in orange and discriminator loss in blue. During the initial epochs, the generator loss exhibits significant oscillations, while the discriminator loss remains nearly constant and close to zero until approximately the 75th epoch. This behavior is consistent with typical GAN training, where the discriminator initially overpowers the generator due to its faster convergence.

At around the 100th epoch, a notable peak in discriminator loss is observed, coinciding with a dip in generator loss. This suggests a momentary rebalancing, as both networks adjust their learning toward a more stable equilibrium [18]. Between the 125th and 200th epochs, the generator loss continues to oscillate within a bounded range of 1 to 5, while the discriminator loss remains spiky but constrained between 0 and 1. From approximately the 250th epoch onward, the generator loss begins to stabilize, indicating that the generator is gradually learning to produce more convincing samples and the adversarial training is reaching a relatively balanced state.

Prior to adopting the ResNet-style architecture described above, we experimented with a baseline generator composed of six conventional upsampling blocks, operating on a 128-dimensional latent vector. The loss function used in this configuration was limited to hinge loss, without any additional structural constraints. However, the results were suboptimal: the generator failed to produce coherent outputs, and the gradients did not converge as expected.

As shown in Figure 2, the generator loss exhibited persistent high-magnitude oscillations, reaching values near 30, and did not demonstrate any convergence throughout training. In contrast, the

discriminator consistently outperformed the generator, indicating a strong imbalance in adversarial training dynamics. This suggests that the discriminator was overfitting early, leaving the generator with limited gradient feedback, and ultimately inhibiting its capacity to learn meaningful representations.

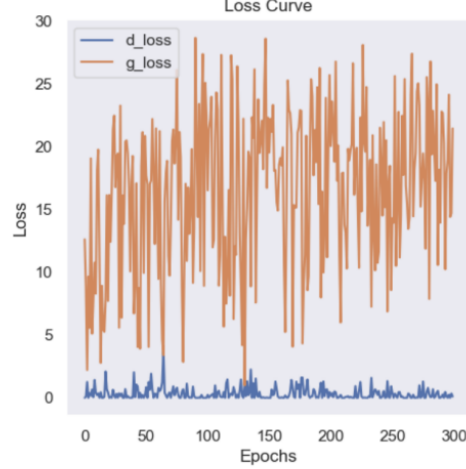


Figure 2: Loss curves of GAN trained with conventional upsampling blocks. The generator failed to converge and the discriminator dominated throughout training.

4.4.2 Training CNN

Given the relatively small dataset size of 2,500 instances (500 samples per class), the base CNN model exhibited a strong tendency to overfit. To mitigate this, early stopping [12] was employed with a variable patience and epochs within a range of 20 to 40. While the training loss consistently decreased in a near-linear fashion to a low value, the validation loss remained unstable, exhibiting spikes throughout training. Notably, the validation loss began to exceed the training loss as early as the 15th epoch as seen in the Figure-3a, signaling the onset of overfitting.

Although early stopping was triggered based on validation loss, the highly fluctuating curve often led to premature termination for lower patience values making the model underfit, or failed to halt training until the final epoch in more lenient settings. Empirically, both training and validation losses were found to be at their lowest around the 19th epoch, beyond which validation loss began to rise. As a result, the baseline model was trained for exactly 19 epochs, and the best-performing weights were saved based on validation loss monitoring.

As shown in Figure 3b, constraining the CNN classifier training, on augmented dataset, to approximately 18 epochs yields the lowest loss and captures the best-performing weights when trained on augmented dataset. This trend is consistent across both training scenarios: using the original dataset and the dataset augmented with synthetic samples.

4.5 Optimization

Several experiments were conducted to stabilize GAN training under the constraint of limited class 4 samples. Achieving a balanced adversarial dynamic requires careful tuning of hyperparameters to prevent the discriminator from overpowering the generator, especially during the early stages of training [18]. Preserving meaningful gradient flow in the generator is critical for effective learning under such conditions.

To address this, multiple regularization strategies were employed on the discriminator. These included L2 weight regularization [11], the introduction of dropout layers [13], a reduced learning

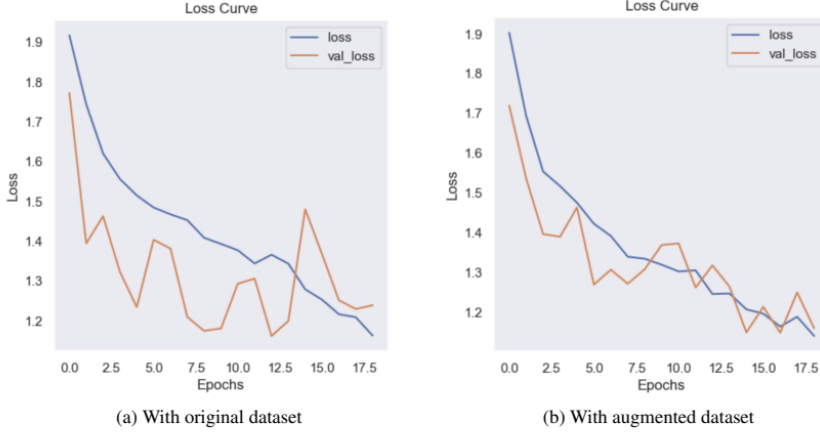


Figure 3: Comparison of training trend of CNN model with and without augmented train dataset

rate [10]. Together, these measures served to attenuate the discriminator’s dominance, allowing the generator more opportunity to improve and guiding the GAN toward a more stable training equilibrium.

Alternative training strategies were explored to address the imbalance between the generator and discriminator during GAN training. One such approach involved conditionally freezing the discriminator for an initial period of 50 epochs, allowing the generator to train without adversarial feedback. This temporarily enabled the generator to produce outputs that could deceive the discriminator; however, once the discriminator resumed training, it rapidly regained dominance, leading to generator degradation. Extending the freeze period to 75 epochs resulted in unstable training dynamics, with the generator loss dropping to negative values and the discriminator loss spiking to extremely high values, reaching nearly 200.

To mitigate this instability, an alternative scheme was tested where the discriminator was trained only every 25 epochs. Nonetheless, the discriminator continued to overpower the generator in subsequent updates, indicating that periodic freezing alone was insufficient to maintain equilibrium. These findings underscore the sensitivity of GAN training to discriminator dynamics and the need for more nuanced balancing strategies.

5 Results

5.1 Quantitative Evaluation

The base CNN model which was trained on both original dataset and augmented dataset achieved the following performance scores

Metric	Original dataset	Synthetic image augmented dataset
Accuracy	0.51	0.54
F1-Score	0.51	0.53
Precision	0.55	0.60
Recall	0.51	0.54

Table 2: Overall performance comparison between baseline and synthetic-augmented models.

The performance metrics are summarized in Table 2. Notably, the model trained on the augmented dataset demonstrated improvements across all metrics. The F1-score increased from 0.51 to 0.53, while precision rose from 0.55 to 0.60, indicating improved confidence in predictions. Recall also

improved from 0.51 to 0.54, reflecting better sensitivity to positive samples. These results suggest that augmenting the training set with synthetic images contributes to modest but consistent gains in overall classification performance.

The following table demonstrates the base CNN model’s performance on both original and synthetic images augmented train dataset pertaining to the class 4 samples:

Metric	Original dataset	Synthetic image augmented dataset
F1-Score	0.12	0.34
Precision	0.14	0.55
Recall	0.10	0.25

Table 3: Performance comparison between baseline and synthetic-augmented models on class 4 samples

To specifically assess the impact on underrepresented classes, we analyzed the performance on class 4 samples. Table 3 shows that the CNN classifier’s ability to correctly identify class 4 instances improved substantially after augmentation. The F1-score increased from 0.12 to 0.34. Precision more than tripled, rising from 0.14 to 0.55, while recall improved from 0.10 to 0.25. These results confirm that synthetic augmentation was particularly effective in improving the classifier’s performance on minority classes, thereby enhancing overall class balance

5.2 Qualitative Comparison of Real and Generated Images

Using the RetinaGAN model trained exclusively on class 4 samples, a total of 500 synthetic class 4 images were generated. To assess the visual and structural fidelity of these images, the Structural Similarity Index Measure (SSIM) score was computed between synthetic and real class 4 images from the training set [23]. The model achieved a maximum SSIM score of approximately 0.80, where a perfect structural match corresponds to a score of 1.0. This indicates that the generated images preserve substantial structural features of the real samples and exhibit high visual similarity.

In addition, the Fréchet Inception Distance (FID) [24,25] was evaluated between the synthetic images and the real class 4 training samples. The model attained a minimum FID score of 88. While lower FID scores in the range of 20–40 are generally indicative of high-fidelity image synthesis, the observed value suggests that the generator has not fully converged and that the realism of the synthetic images can be further improved. These results highlight the potential of the model while also indicating room for refinement in learning finer distributional characteristics of the real data.

The following figure displays the class 4 images from tain dataset.

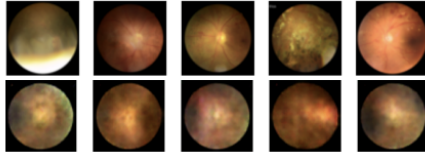


Figure 4: Visual representation of real class 4 (top) and synthetic retina images (bottom).

Visually, the synthetic images shown in the bottom row of Figure 4 appear largely indistinguishable from the real images in the top row at first glance. However, upon closer inspection, the generated samples exhibit a degree of blurriness and lack sharp anatomical details, particularly in the definition of blood vessels and the foveal region. This degradation in fine-grained structure is likely attributable to limitations in the generator’s capacity to capture high-frequency components of the target distribution, a common challenge in GAN-based medical image synthesis.

6 Discussion

Initially, we employed Binary Cross-Entropy (BCE) as the loss function. However, BCE led to an overconfident discriminator early in training, which hindered the generator’s ability to improve. While the generated images were structurally coherent, they exhibited significant blurriness and lacked photorealistic fidelity. To address this imbalance and reduce discriminator overfitting, we adopted a hinge loss formulation. This provided more stable gradients and allowed the generator to continue learning even when the discriminator was dominant. Although hinge loss improved structural realism, fine-grained retinal features—such as the fovea and vascular textures—remained underrepresented. To further enhance perceptual quality, we introduced a regularization term into the generator loss by penalizing the Structural Similarity Index (SSIM), scaled by a factor of 0.3. This addition encouraged the generator to prioritize spatial coherence, resulting in sharper visual representations that captured subtle anatomical details of the retina. Incorporating perceptual loss into the generator was also explored; however, due to its disproportionately high magnitude, the generator loss failed to converge and exhibited significant oscillations, occasionally reaching values as high as 25.

When the base CNN model was trained on the augmented dataset, a noticeable shift in classification behavior was observed, particularly affecting classes 1 and 3. We attribute this to two primary factors: (1) retinal images across different classes exhibit high structural similarity and complex color variation, while the limited number of samples per class constrains the model’s ability to generalize; and (2) the GAN-generated images for class 4 did not accurately replicate fine anatomical details. These synthetic samples were often blurry and lacked critical retinal features such as vessel patterns and macular structure. Due to this visual overlap with other classes and insufficient detail in the augmented images, the CNN classifier exhibited an increase performance on classes 1 and 3 when trained on the augmented dataset.

7 Conclusion and Future work

In this work, we employed a GAN architecture with ResNet-style blocks to generate synthetic images for data augmentation, addressing the severe class imbalance present in the dataset. A VGG19-based CNN classifier was used to evaluate model performance on both the original and augmented datasets. Results demonstrate that the inclusion of synthetic images improves classification metrics and enhances generalization, particularly for underrepresented classes.

While the generated images exhibit strong structural similarity to real samples, they lack fine-grained anatomical detail, resulting in some visual blurriness. This suggests that the current model has not fully captured high-frequency features and could benefit from further hyperparameter tuning. However, identifying an optimal hyperparameter configuration is non-trivial; improper settings can lead to vanishing gradients in the generator or allow the discriminator to dominate, destabilizing training.

Overall, our findings highlight the effectiveness of GAN-based augmentation in mitigating class imbalance. In future work, we plan to investigate alternative GAN architectures such as Wasserstein GAN (WGAN) and PatchGAN, with the goal of improving perceptual quality and achieving lower FID scores and higher SSIM scores.

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